THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 7 — MATHS · NCERT-TYPE CHAPTER TEST SET 2

Chapter 7: **A Tale of Three Intersecting Lines** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The exterior angle of a triangle is $(3x + 20)^\circ$ and the interior opposite angles are 80° and 45°. The value of x is: 1

(a) 34 **(b)** 35 **(c)** 36 **(d)** 37

2. If $\triangle ABC \cong \triangle PQR$, then $\angle B$ equals: 1

(a) $\angle Q$ **(b)** $\angle A$ **(c)** $\angle R$ **(d)** $\angle P$

3. In $\triangle PQR$, the angle included between sides PQ and QR is: 1

(a) $\angle Q$ **(b)** $\angle P$ **(c)** $\angle R$ **(d)** none of these

4. Which set of angles can be the angles of a triangle? 1

(a) $40^\circ, 40^\circ, 110^\circ$ **(b)** $35^\circ, 45^\circ, 90^\circ$ **(c)** $60^\circ, 60^\circ, 70^\circ$ **(d)** $35^\circ, 45^\circ, 100^\circ$

5. In $\triangle PQR$, $PQ = PR$ and $\angle Q = 50^\circ$. Then $\angle R$ is: 1

(a) 40° **(b)** 130° **(c)** 80° **(d)** 50°

6. Two triangles have angles $40^\circ, 70^\circ, 70^\circ$ each. They are: 1

(a) not necessarily congruent **(b)** never congruent
(c) always congruent **(d)** congruent by SSS

7. Two sides of a triangle are 4 cm and 10 cm. The third side could be: 1

(a) 6 cm **(b)** 8 cm **(c)** 14 cm **(d)** 16 cm

8. The angles of a triangle are x°, $2x^\circ$ and $2x^\circ$. The value of x is: 1

(a) 35 **(b)** 36 **(c)** 37 **(d)** 38

9. The exterior angle of a triangle is $(3x + 5)^\circ$ and the interior opposite angles are 45° and 35°. The value of x is: 1

(a) 26 **(b)** 24 **(c)** 25 **(d)** 27

10. If $\triangle ABC \cong \triangle PQR$, then side AB equals: 1

(a) QR **(b)** PQ **(c)** AC **(d)** PR

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. The angles of a triangle are $(2x + 18)^\circ$, $(3x - 12)^\circ$ and $(x + 6)^\circ$. Find x and the three angles. 2

12. In $\triangle ABC$, side BC is extended to D, $\angle ACD = (5x - 6)^\circ$, $\angle A = (2x + 14)^\circ$ and $\angle B = (x + 36)^\circ$. Find x and $\angle ACD$. 2

13. Two sides of a triangle are 8 cm and 11 cm. Find the least and the greatest whole-number length of the third side. 2

14. The angles of a triangle are $(2x + 20)^\circ$, $(3x - 5)^\circ$ and $(x + 15)^\circ$. Find x and the three angles. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In isosceles $\triangle ABC$ with $AB = AC$, $\angle A = (x + 30)^\circ$ and $\angle B = (2x - 25)^\circ$. Find x and all three angles. 3

16. In $\triangle ABC$ and $\triangle PQR$, $AB = PQ = 8$ cm, $BC = QR = 10$ cm and $\angle B = \angle Q = 65^\circ$. Name the congruence rule, and find PR if $AC = 15$ cm. Find also the perimeter of $\triangle PQR$. 3

17. The exterior angle at C of $\triangle ABC$ is 130° and $\angle A : \angle B = 6 : 7$. Find all three angles of the triangle. 3

OR

In isosceles $\triangle ABC$ with $AB = AC$, $\angle A = (x + 15)^\circ$ and $\angle B = (2x - 5)^\circ$. Find x and all three angles. 3

18. In $\triangle ABC$ and $\triangle PQR$, $AB = PQ = 4$ cm, $BC = QR = 12$ cm and $\angle B = \angle Q = 70^\circ$. Name the congruence rule, and find PR if $AC = 13$ cm. Find also the perimeter of $\triangle PQR$. 3

OR

The exterior angle at C of $\triangle ABC$ is 150° and $\angle A : \angle B = 2 : 3$. Find all three angles of the triangle. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. The angles of a triangle are $(x + 30)^\circ$, $(x - 10)^\circ$ and $(2x + 20)^\circ$. Find x, the three angles and classify the triangle. Also find the three exterior angles (one at each vertex). 5

OR

Two sides of a triangle are 8 cm and 8 cm. 5
(i) Between which two lengths does the third side lie?
(ii) How many whole-number lengths are possible?
(iii) Find the smallest and the largest possible perimeter in whole centimetres.

20. The angles of a triangle are $(x + 40)^\circ$, $(x - 8)^\circ$ and $(2x + 12)^\circ$. Find x, the three angles and classify the triangle. Also find the three exterior angles (one at each vertex). 5

OR

Two sides of a triangle are 6 cm and 15 cm. 5
(i) Between which two lengths does the third side lie?
(ii) How many whole-number lengths are possible?
(iii) Find the smallest and the largest possible perimeter in whole centimetres.

END OF QUESTION PAPER — **Total: 40 Marks**